Logistic Regression; Probability

Jericho Lawson

MATH 240

Week 7, Lecture #1 — Ch. 9

Reminders

- Homework #5 due on Friday at 11:59pm
  - One file for textbook problems
  - One file for R code
- No more checkpoints and lab assignments up until fall break!

Revised Steps for Logistic Regression

We can do the following tasks when performing logistic regression:

1. **Fitting**
2. **Classification**

Tutorial: Iris Dataset - Introduction

- Measurements of sepal length, sepal width, petal length, and petal width for 50 flowers from each of 3 species of iris
- The species are setosa, versicolor, and virginica
- For logistic regression, we will just focus on two species: setosa and versicolor
- Number of observations $n = 100$; number of variables $k = 5$

Tutorial: Iris Dataset - Introduction

- `data(iris)`
- `flowers = iris[1:100, ]`
- `head(flowers)`

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.10	3.50	1.40	0.20	setosa
2	4.90	3.00	1.40	0.20	setosa
3	4.70	3.20	1.30	0.20	setosa
4	4.60	3.10	1.50	0.20	setosa
5	5.00	3.60	1.40	0.20	setosa
6	5.40	3.90	1.70	0.40	setosa

Tutorial: Iris Dataset - Introduction

Main question to ask:

- How does the sepal length affect whether we classify a flower as setosa or versicolor?

With the given variables, we will determine which variables affect the classification of a flower (and to what extent).

Fitting

For logistic regression, we now use the `glm()` function to fit a model. The `glm()` is in the following for for a logistic regression model:

- `glm(y ~ x1 + x2 + ... + xp, data, family = "binomial")`

- **Important**: The `family = "binomial"` argument specifies we want to fit a logistic regression model out of all possible GLMs

The code will find the most appropriate β coefficients for a logistic regression model:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k$$

Tutorial: Iris Dataset - Fitting

Suppose we want to construct a model that determines whether we classify a flower as setosa or versicolor based on the sepal length (x_1), sepal width (x_2), petal length (x_3), and petal width (x_4)?

As such, our logistic regression model will be the following:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

Tutorial: Iris Dataset - Fitting

We can now fit the model and get a summary of the model as such:

- `mod = glm(Species ~ Sepal.Length, data = flowers, family = "binomial")`
- `summary(mod)`

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-27.8315	5.4336	-5.12	0.0000
Sepal.Length	5.1403	1.0066	5.11	0.0000

Tutorial: Iris Dataset - Interpretation

Using the information from the `summary()` command (where $x_1 = x_{\text{Sepal.Length}}$), we can write the logistic regression model as:

$$\begin{aligned}
\log\left(\frac{\hat{p}}{1-\hat{p}}\right) &= \hat{\beta}_0 + \hat{\beta}_1 x_1 \\
&= -27.8315 + 5.1403 x_1 \\
\hat{p} &= \frac{e^{-27.8315+5.1403x_1}}{1+e^{-27.8315+5.1403x_1}}
\end{aligned}$$

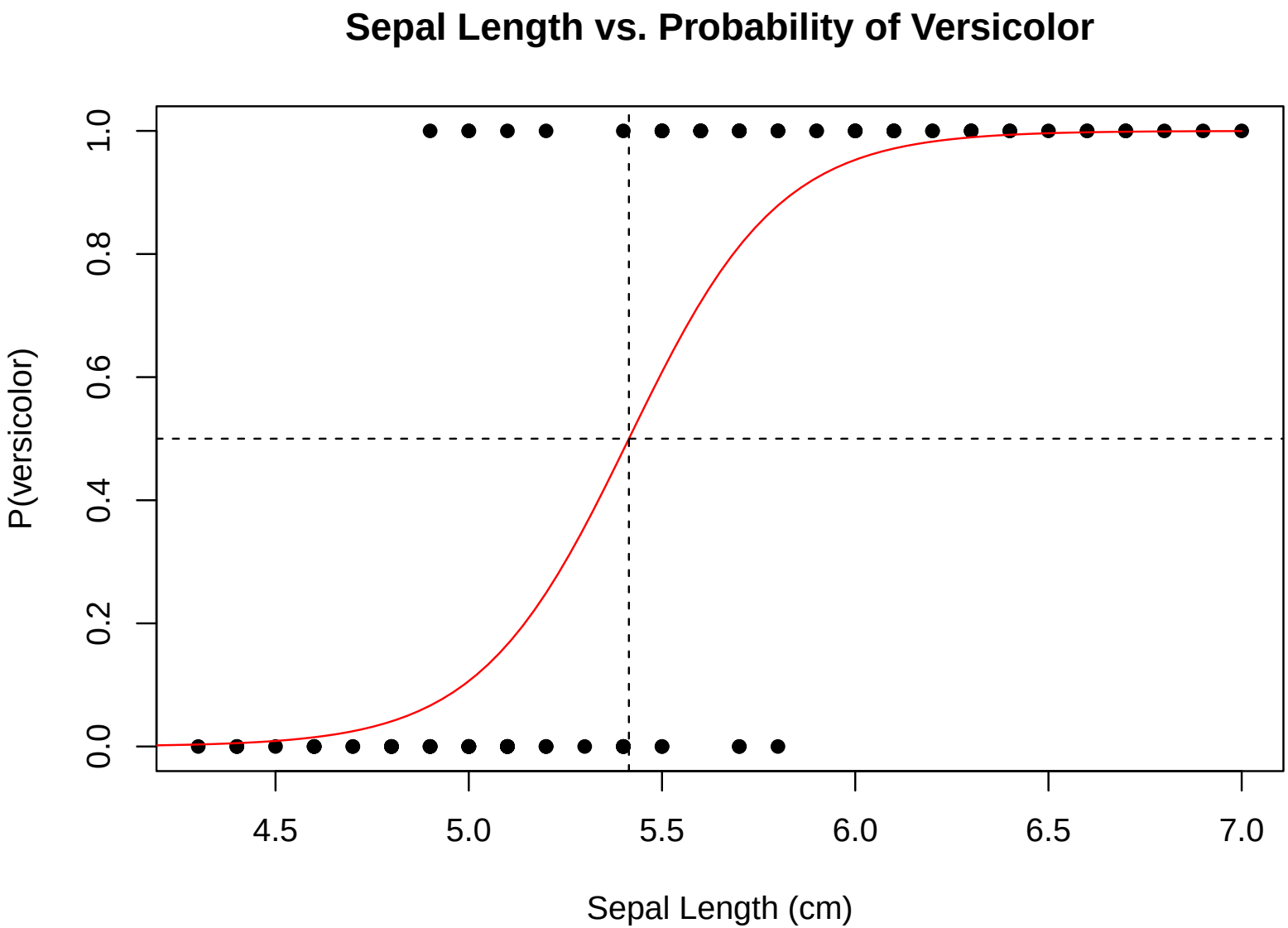
Tutorial: Iris Dataset - Interpretation

We can calculate the estimated probability of a flower being versicolor for particular values of the sepal length using the following formula for $\hat{p}$:

$$\hat{p} = \frac{e^{-27.8315+5.1403x_1}}{1 + e^{-27.8315+5.1403x_1}}$$

Sepal Length	4	4.5	5	5.5	6	6.5	7
$P(versicolor)$	0.0007	0.0090	0.1062	0.6083	0.9530	0.9962	0.9997

Tutorial: Iris Dataset - Interpretation



Classification

Whenever we want to classify new points based on our logistic regression model, we will need to calculate the probabilities using the best-fit line and determine a threshold for classification (typically 0.5).

Recall that if we want to calculate a probability manually for some observation with predictor values $X_1 = x_1, X_2 = x_2, \ldots, X_p = x_p$, we can use the following formula:

$$\hat{p} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_k x_k}}$$

Classification

For some threshold τ, we can classify an observation as one of the two categories:

$$\hat{y} = \begin{cases} 1, \text{ level of interest} & \hat{p} \geq \tau \\ 0, \text{ reference level} & \hat{p} < \tau \end{cases}$$

In most cases (and in this class), we set $\tau = 0.5$. However, other values for τ can be used on a case-by-case basis.

- Note: The reference level will be defined as 0 for your response. The level of interest is the same level for which you are calculating $\hat{p}$ for.

Classification with R

Suppose we have the following sepal lengths of 10 new flowers:

- 6.3, 5.8, 7.1, 6.3, 6.5, 7.6, 4.9, 7.3, 6.7, 7.2

We can use the following code to get new predictions for y:

- `new_data <- data.frame(Sepal.Length = c(6.3, 5.8, 7.1, 6.3, 6.5, 7.6, 4.9, 7.3, 6.7, 7.2))`
- `probs = predict(mod, new_data, type = "response")`
- `pred = as.numeric(probs >= 0.5)`

```
[1] 1 1 1 1 1 1 0 1 1 1
```

Probability - What is it?

- "logic of uncertainty" (and variability)
- Building tools to describe and understand apparent randomness
  - From sampling, distributions, and testing
  - e.g. p-value, probability of event occurring, proportions

Probability - What is it?

Questions where probability is needed:

- If I flip a coin once, what is the chance it'll land on heads?
- What is the chance it'll rain in the next 24 hours?
- What is the chance that some statistic will be more extreme than some value?

Language of Probability

We need three components:

1. A random experiment
   - Some activity, process, or (actual or thought) experiment whose outcome is uncertain
   - e.g. rolling a die
2. A sample space
   - A set/collection of all possible outcomes from the random experiment, denoted as $\mathcal{S}$
   - e.g. $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ for rolling a die
3. An event
   - A collection of outcomes
   - e.g. Getting an odd number on a roll of a die

Language of Probability

- Can be done theoretically or empirically
  - Theoretically: use of formulas and counting to identify the correct probabilities
    - ★ Counting: Count up number of outcomes for some event (use of tree maps, Venn diagrams)
  - Empirically: use of experiments to estimate true probability of an event occurring
    - ★ e.g. law of large numbers ($\hat{p}_n \to p$)

Activity: Components in Random Experiments

For each of these scenarios, determine the **sample space** and an **event** from the sample space.

1. Dealing a card from a set of playing cards
2. Analyzing the number of classes a student is currently taking
3. Playing a single turn in Monopoly (or any other board game)